



Time Series Analysis

Seminar 6: VAR models and Granger causality



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ CCF: Cross-Correlation Function
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ FEVD: Forecast Error Variance Decomposition
- ▣ GDP: Gross Domestic Product
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ HQ: Hannan–Quinn (information criterion)
- ▣ OLS: Ordinary Least Squares
- ▣ ROBOR: Romanian Interbank Offer Rate
- ▣ SE: Standard Error
- ▣ VAR: Vector AutoRegression

Today's question and route

- **Question:** when we model several series together, which of them helps to forecast the others, and how does a shock in one spread to the rest?
 - ▶ this seminar comes **before** Lecture 6: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: a VAR(1) by hand (stability, mean, forecasts); impulse responses with the Cholesky factor; a variance decomposition; a Granger F test; lag selection
 - ▶ Part B: does the S&P 500 lead the BET (daily and weekly)? a VAR for Romanian GDP growth, inflation and ROBOR 3M; the EUR/RON rate added to it
 - ▶ Part C: an open question on the “price puzzle” and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	a VAR(1) by hand: stability, mean, forecasts	Solved, Proposed	A1
A3, A4	orthogonalised impulse responses; a variance decomposition	Solved, Proposed	A3
A5, A6	a Granger F test; lag selection by AIC, BIC and HQ	Solved, Proposed	A5
B1, B2	S&P 500 and BET: daily; weekly	Solved, Proposed	B1
B3, B4	a Romanian macro VAR; the EUR/RON rate added	Solved, Proposed	B3
C1, C2	the price puzzle; what is wrong in an AI answer?	Proposed	B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
S&P 500, BET	EODHD	daily, close	2000–2026
Real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, seasonally adjusted	2005–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2004–2026
ROBOR 3M	Eurostat (irt_st_m)	monthly, % per year	2005–2026
EUR/RON	BNR reference rate	daily	2005–2026

- Quarterly VAR variables: $g_t = 100\Delta \ln \text{GDP}_t$; π_t : 12-month HICP inflation (quarterly mean); i_t : ROBOR 3M (quarterly mean); s_t : $100\Delta \ln$ of the quarterly mean EUR/RON
- In the notebook: VAR from statsmodels, granger_F, ic_table, girf; no account or key is needed

Prerequisites for Today (1/5): the VAR model and its stability

- ▣ **VAR(p)**: each variable depends on the last p values of all variables
 - ▶ $Y_t = c + A_1 Y_{t-1} + \dots + A_p Y_{t-p} + \varepsilon_t$
 - ▶ Y_t : the vector of the K series at date t ; c : the $K \times 1$ vector of constants; A_j : $K \times K$ coefficient matrices
 - ▶ ε_t : white-noise shocks with covariance matrix $E[\varepsilon_t \varepsilon_t'] = \Sigma$; the prime $'$ denotes the transpose
 - ▶ equation by equation: a regression of each variable on a constant and the p lags of all variables; a_{jk} : the effect of $y_{k,t-1}$ on y_{jt}
- ▣ **Stability** of a VAR(1): all eigenvalues λ of A have $|\lambda| < 1$, so the effect of a shock dies out
 - ▶ for $K = 2$: $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$, with the trace $\text{tr}(A) = a_{11} + a_{22}$ and the determinant $\det(A) = a_{11}a_{22} - a_{12}a_{21}$
 - ▶ complex $\lambda = a \pm bi$: modulus $|\lambda| = \sqrt{a^2 + b^2}$; VAR(p): the eigenvalues of the companion matrix $\begin{pmatrix} A_1 & A_2 \\ I & 0 \end{pmatrix}$ (for $p = 2$; I : the identity matrix)

Prerequisites for Today (2/5): mean, forecasts and estimation

- **Mean** of a stable VAR(1): the level to which the series return, $\mu = (I - A)^{-1}c$
 - ▶ inverse of a 2×2 matrix: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
- **Forecasts** by recursion: $\hat{Y}_{T+1} = c + AY_T$, $\hat{Y}_{T+2} = c + A\hat{Y}_{T+1}$
 - ▶ Y_T : the last observation; $\hat{Y}_{T+h}$: the forecast h periods ahead; the shocks are replaced by their mean, zero
- **Estimation**: OLS equation by equation (all equations have the same regressors); $K + pK^2$ coefficients
 - ▶ the variables must be stationary ([Chapter 3](#)): growth rates, returns, or rates that do not trend

Prerequisites for Today (3/5): lag order and diagnostics

- ▣ **Lag order:** an information criterion trades fit against the number of coefficients; the smallest value is best
 - ▶ $IC(p) = \ln \det \tilde{\Sigma}(p) + c_T pK^2/T$, computed for every p on a common sample
 - ▶ $\tilde{\Sigma}(p)$: the residual covariance matrix of the VAR(p), divided by T ; T : the number of observations; pK^2 : the number of lag coefficients
 - ▶ $c_T = 2$ (AIC), $\ln T$ (BIC), $2 \ln \ln T$ (HQ); BIC chooses the smallest p , AIC the largest
- ▣ **Portmanteau test:** H_0 : residuals without auto- and cross-correlation up to lag h
 - ▶ statistic Q_h ; under H_0 , $Q_h \sim \chi^2(K^2(h-p))$
 - ▶ a small p-value: add lags or variables; normality (Jarque–Bera) often fails because of crises

Prerequisites for Today (4/5): Granger causality

- ▣ x **Granger-causes** y if the past of x improves the forecast of y beyond the past of y and of the other variables
 - ▶ in a VAR(p): H_0 : the p coefficients of $x_{t-1}, \dots, x_{t-p}$ in the equation of y are all zero
- ▣ F **test**: does dropping the lags of x raise the residual sum of squares by much?
 - ▶ $F = \frac{(RSS_R - RSS_U)/q}{RSS_U/(T - k)}$
 - ▶ RSS_U, RSS_R : residual sums of squares of the equation of y with and without the lags of x ; q : number of restrictions; k : coefficients of the unrestricted equation
 - ▶ reject H_0 if F exceeds the critical value of $F(q, T - k)$ (about 3.1 for $q = 2$ and $T - k \approx 80$ –100)
- ▣ **Not** cause and effect: a statement about predictability, relative to the chosen variables
 - ▶ pitfalls: an omitted common driver; effects within the same period (instantaneous causality, in Σ); non-stationary series

Prerequisites for Today (5/5): impulse responses and FEVD

- **Impulse responses** of a VAR(1): $\Phi_h = A^h$; column k : the path of all variables h periods after a unit shock to ε_k
 - ▶ the shocks are correlated, so we write $\varepsilon_t = P u_t$ with uncorrelated, unit-variance shocks u_t and $\Sigma = PP'$ (**Cholesky**, P lower triangular)
 - ▶ $\Theta_h = \Phi_h P$; element $\theta_{jk,h}$: the response of y_j after h periods to a shock u_k of one standard deviation
 - ▶ 2×2 : $p_{11} = \sqrt{\sigma_{11}}$, $p_{21} = \sigma_{21}/p_{11}$, $p_{22} = \sqrt{\sigma_{22} - p_{21}^2}$; σ_{jk} : the elements of Σ
- **The ordering matters**: the first variable does not react within the period to the shocks of the others
 - ▶ the **generalised** responses [Pesaran and Shin \(1998\)](#) do not depend on the order: $\Phi_h \Sigma e_k / \sqrt{\sigma_{kk}}$, e_k : column k of I
- **FEVD** (forecast error variance decomposition): the share of shock k in the h -step forecast error variance of y_j
 - ▶ $\omega_{jk}(h) = \sum_{s < h} \theta_{jk,s}^2 / \sum_{s < h} \sum_m \theta_{jm,s}^2$; each row sums to 100%

A1: a VAR(1) by hand [Solved]

Task

- $c = (0.4, 0.8)'$, $A = \begin{pmatrix} 0.6 & 0.2 \\ 0.2 & 0.6 \end{pmatrix}$; last observation $Y_T = (3, 2)'$.
- 1. Write the two equations.
- 2. Find the eigenvalues of A and decide whether the VAR is stable.
- 3. Compute the mean μ .
- 4. Compute $\hat{Y}_{T+1}$ and $\hat{Y}_{T+2}$.
- Report: two equations, two eigenvalues, μ , two forecasts.

Solution

- 1. $y_{1t} = 0.4 + 0.6y_{1,t-1} + 0.2y_{2,t-1} + \varepsilon_{1t}$;
 $y_{2t} = 0.8 + 0.2y_{1,t-1} + 0.6y_{2,t-1} + \varepsilon_{2t}$
- 2. $\text{tr} = 1.2$, $\det = 0.36 - 0.04 = 0.32$: $\lambda^2 - 1.2\lambda + 0.32 = 0$,
 $\lambda = 0.8$ and 0.4 ; both below 1: stable
- 3. $I - A = \begin{pmatrix} 0.4 & -0.2 \\ -0.2 & 0.4 \end{pmatrix}$, $\det = 0.12$;
 $\mu = \frac{1}{0.12} \begin{pmatrix} 0.4 & 0.2 \\ 0.2 & 0.4 \end{pmatrix} \begin{pmatrix} 0.4 \\ 0.8 \end{pmatrix} = \begin{pmatrix} 2.667 \\ 3.333 \end{pmatrix}$
- 4. $\hat{Y}_{T+1} = (0.4 + 1.8 + 0.4, 0.8 + 0.6 + 1.2)' = \begin{pmatrix} 2.60 \\ 2.60 \end{pmatrix}$;
 $\hat{Y}_{T+2} = \begin{pmatrix} 2.48 \\ 2.88 \end{pmatrix}$: moving towards μ

A2: stable or not? [Proposed]

Solution discussed in class

Task

- ▣ Two VAR(1) matrices: (a) $A = \begin{pmatrix} 0.9 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}$; (b) $A = \begin{pmatrix} 0.5 & -0.4 \\ 0.4 & 0.5 \end{pmatrix}$. Model: A1.
- ▣ 1. For each matrix, compute the trace, the determinant and the eigenvalues.
- ▣ 2. Decide whether each VAR is stable.
- ▣ 3. Describe how the responses to a shock behave in each case.
- ▣ Report: four eigenvalues (or moduli), two decisions and two sentences.



A3: orthogonalised impulse responses [Solved]

Task

- ▣ The VAR(1) of A1 with $\Sigma = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix}$, order (y_1, y_2) .
- ▣ 1. Compute the Cholesky factor P.
- ▣ 2. Compute $\Theta_0 = P$ and $\Theta_1 = AP$, and $\Phi_2 = A^2$.
- ▣ 3. Compute P again with the order (y_2, y_1) and say what changes on impact.
- ▣ Report: three matrices and one sentence.

Solution

- ▣ 1. $p_{11} = 1$, $p_{21} = 0.4$, $p_{22} = \sqrt{1 - 0.16} = 0.917$:

$$P = \begin{pmatrix} 1.000 & 0.000 \\ 0.400 & 0.917 \end{pmatrix}$$
- ▣ 2. $\Theta_1 = \begin{pmatrix} 0.680 & 0.183 \\ 0.440 & 0.550 \end{pmatrix}$; $\Phi_2 = \begin{pmatrix} 0.40 & 0.24 \\ 0.24 & 0.40 \end{pmatrix}$; after $u_1 = 1$: y_2 moves by 0.4 at once and by 0.44 one period later
- ▣ 3. Order (y_2, y_1) : $P = \begin{pmatrix} 0.917 & 0.400 \\ 0.000 & 1.000 \end{pmatrix}$ (in the original order); now y_1 reacts on impact to the y_2 shock (0.4) and y_2 does not react to the y_1 shock

A4: a variance decomposition by hand [Proposed]

Solution discussed in class

Task

- ▣ Use Θ_0 and Θ_1 of A3 (order y_1, y_2). Model: A3.
- ▣ 1. Compute the share of u_1 in the 1-step forecast error variance of y_2 .
- ▣ 2. Compute the 2-step forecast error variance of y_2 and the share of u_1 in it.
- ▣ 3. Compute the share of u_2 in the 2-step error variance of y_1 , and explain why it is 0 at one step.
- ▣ Report: three percentages and one sentence.

A5: a Granger F test [Solved]

Task

- ▣ A bivariate VAR(2) on $T = 100$ observations. In the equation of y_1 : $RSS_U = 45.2$ (with the two lags of y_2) and $RSS_R = 52.8$ (without them).
- ▣ 1. Give H_0 , the number of restrictions q and the number of coefficients k of the unrestricted equation.
- ▣ 2. Compute F .
- ▣ 3. Decide at 5% and state the conclusion in words.
- ▣ Report: H_0 , F , a decision and one sentence.

Solution

- ▣ 1. $H_0: a_{12}^{(1)} = a_{12}^{(2)} = 0$ (y_2 does not Granger-cause y_1); $q = 2$;
 $k = 1 + 2 \cdot 2 = 5$
- ▣ 2. $F = \frac{(52.8 - 45.2)/2}{45.2/95} = \frac{3.8}{0.476} = 7.99$
- ▣ 3. $7.99 > 3.09$, the 5% value of $F(2, 95)$ ($p < 0.001$): reject;
the past of y_2 helps to forecast y_1 , which does not prove that y_2 causes y_1

A6: choosing p by hand [Proposed]

Solution discussed in class

Task

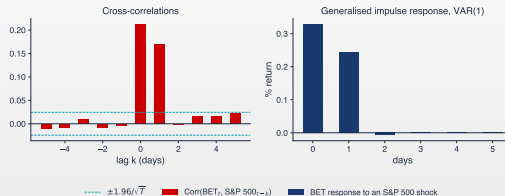
- ▣ A VAR with $K = 3$ on a common sample of $T = 80$ quarters gives $\ln \det \tilde{\Sigma}(p) = 1.50; 0.60; 0.25; 0.00; -0.15$ for $p = 0, \dots, 4$. Model: A5.
- ▣ 1. Compute the penalty per lag $c_T K^2 / T$ for AIC, BIC and HQ.
- ▣ 2. Compute the three criteria for each p and find the minima.
- ▣ 3. Say which p you would start from with 80 quarters, and why.
- ▣ Report: three penalties, a table of 15 values, three orders and one sentence.



B1: does the S&P 500 lead the BET? [Solved]

- ▣ **Question:** do yesterday's S&P 500 returns help to forecast today's BET returns, and the other way round?
- ▣ **Inputs:** daily log returns (%) of the S&P 500 and the BET since 2000, on common trading days
- ▣ **Tasks**
 1. Compute the cross-correlations $\text{Corr}(\text{BET}_t, \text{S\&P}_{t-k})$ for $k = -5, \dots, 5$.
 2. Choose p for a bivariate VAR with AIC, BIC and HQ (maximum 10) and estimate the VAR with the HQ order.
 3. Test Granger causality in both directions.
 4. Compute the generalised response of the BET to an S&P 500 shock for 5 days.
 5. Interpretation: why does New York lead Bucharest by one day?
- ▣ **Report:** three cross-correlations, the chosen p , two F tests, two responses and one sentence
- ▣ **Notebook:** section B1

B1: solution [Solved]



- $T = 6466$ days; CCF: 0.21 ($k = 0$), 0.170 ($k = 1$), -0.003 ($k = -1$); orders: AIC 9, BIC 1, HQ 1: VAR(1)
- S&P 500 $\rightarrow$ BET: $F = 156.2$ ($p < 0.001$); BET $\rightarrow$ S&P 500: $F = 2.25$ ($p = 0.133$); coefficient of $\text{S\&P}_{t-1}$: 0.182 (SE 0.015)
- Generalised response of the BET: 0.33% on the same day, 0.24% the next day, -0.007% after two days
- Interpretation: Wall Street closes after Bucharest, so news of the American afternoon reaches the BET only the next day: a time-zone effect, not a slow market

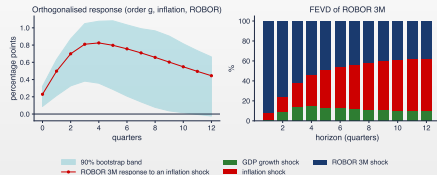
B2: the same question with weekly returns [Proposed]

- **Question:** does the lead of the S&P 500 survive when we use Friday-to-Friday weekly returns?
- **Inputs:** weekly log returns (%) of the S&P 500 and the BET from the Friday closes; model: B1
- **Tasks**
 1. Repeat steps 1–4 of B1 on weekly returns.
 2. Compare the lag-1 cross-correlation and the F statistic with the daily ones.
 3. Interpretation: why is the lead much weaker at the weekly frequency?
- **Report:** the same items as in B1 and two sentences
- **Notebook:** section B2

B3: a VAR for the Romanian economy [Solved]

- **Question:** how do GDP growth, inflation and the 3-month interest rate interact in Romania?
- **Inputs:** $Y_t = (g_t, \pi_t, i_t)'$, quarterly, 2005Q1–2026Q2
- **Tasks**
 1. Choose p with AIC, BIC and HQ ($p \leq 6$) and estimate a VAR(2).
 2. Check stability (largest modulus of the companion eigenvalues) and run the portmanteau test with $h = 8$ and $h = 12$.
 3. Run the six Granger F tests.
 4. Plot the orthogonalised response of ROBOR to an inflation shock (order g, π, i) and the FEVD of ROBOR.
 5. Interpretation: do the results show that the central bank reacts to inflation?
- **Report:** three orders, one modulus, two portmanteau tests, a table of six tests, the chart and one sentence
- **Notebook:** section B3

B3: solution [Solved]



- ▣ Orders: AIC 5, BIC 1, HQ 2; VAR(2) on 84 quarters; largest modulus 0.869: stable; $Q_8 = 76.7$ (p 0.023), $Q_{12} = 105.3$ (p 0.130)
- ▣ Granger (p): $g \rightarrow i$ 0.001, $\pi \rightarrow i$ 0.011, $i \rightarrow g$ 0.022, $\pi \rightarrow g$ 0.059, $i \rightarrow \pi$ 0.417, $g \rightarrow \pi$ 0.853
- ▣ ROBOR after an inflation shock: 0.23 pp on impact, 0.82 pp after 4 quarters; FEVD of ROBOR at 12 quarters: inflation 53%, own 37%
- ▣ Interpretation: they show that ROBOR follows inflation, as a reaction function would imply; ROBOR is a market rate, and the VAR cannot separate the BNR's decisions from market expectations

B4: adding the EUR/RON exchange rate [Proposed]

- **Question:** does the depreciation of the leu help to forecast inflation (exchange-rate pass-through)?
- **Inputs:** $Y_t = (g_t, s_t, \pi_t, i_t)'$ with s_t the quarterly EUR/RON change, from 2005Q4; model: B3
- **Tasks**
 1. Choose p with the three criteria ($p \leq 4$) and estimate a VAR(2).
 2. Test whether s Granger-causes π , i and g , and whether i and π Granger-cause s .
 3. Compute the share of EUR/RON shocks in the FEVD of inflation at 1, 4, 8 and 12 quarters.
 4. Interpretation: does the result mean that the exchange rate does not affect Romanian prices?
- **Report:** three orders, five F tests, four shares and two sentences
- **Notebook:** section B4

C1: the price puzzle [Proposed]

- **Question:** why does inflation rise after a ROBOR shock in the VAR of B3, and does it change when the exchange rate or more lags are added?
- **Inputs:** the VARs of B3 and B4; ROBOR ordered last; models: B3, B4
- **Tasks**
 1. Plot the response of inflation to a ROBOR shock in the VAR(2) of B3, the VAR(2) of B4 and a VAR(4) with the four variables of B4.
 2. Report the largest and the smallest response and their horizons.
 3. List two pieces of information that the central bank has and the VAR does not.
 4. Interpretation: is a positive response of inflation a proof that higher rates raise prices?
- **Report:** the chart, six numbers and a project plan
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to analyse the VAR of B3. The answer:
- ▣ (a) AIC chooses $p = 5$, so the true lag order of the economy is 5.
- ▣ (b) The largest eigenvalue modulus of A1 is 1.37, so the VAR(2) is explosive.
- ▣ (c) The portmanteau test at 8 lags has $p = 0.023$, so the residuals are white noise.
- ▣ (d) GDP growth Granger-causes ROBOR, so faster growth causes the BNR to raise rates.
- ▣ (e) Cholesky impulse responses do not depend on the order of the variables.
- ▣ (f) In each row of the FEVD the shares add up to 100%.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- ▣ Stability: eigenvalues of A (or of the companion matrix) inside the unit circle; then forecasts converge to μ
- ▣ Granger F test: compare the equation with and without the lags of the candidate variable
- ▣ Cholesky responses depend on the ordering; the generalised responses do not
- ▣ S&P 500 leads the BET by one day because of the time zones; ROBOR follows Romanian inflation
- ▣ An AI answer is a draft: check the direction of each claim and which matrix decides stability

After the seminar

- Lecture 6 develops each topic of today: vector processes, the companion form, lag selection, diagnostics, Granger causality and its pitfalls, impulse responses, FEVD, spillovers, forecasting, structural VARs
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: monetary policy shocks in Romania with an extended VAR
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 7; [Hyndman and Athanasopoulos \(2021\)](#), Sec. 12.3; [Stock and Watson \(2001\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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